

Math 344, S19
Exam 1

Name: *Answer*

Instructions:

- There are totally 5 problems.
- You must show all of your work to receive a credit for a correct answer.
- No graphical calculator is allowed in the quiz and exam.

Problem	Points	Score
1	20	
2	20	
3	22	
4	20	
5	18	
Total	100	

Good Luck !

Problem 1. (20 points) Given two vectors in the three dimensional space $\mathbf{v} = (3, \sqrt{3}, 2\sqrt{6})$ and $\mathbf{w} = (0, -\frac{4\sqrt{3}}{3}, \frac{4\sqrt{6}}{3})$.

(a) Find the length of $\mathbf{v}$ and $\mathbf{w}$.

$$\|\mathbf{v}\| = \sqrt{3^2 + (\sqrt{3})^2 + (2\sqrt{6})^2} = \sqrt{9 + 3 + 24} = \sqrt{36} = 6$$

$$\|\mathbf{w}\| = \sqrt{0^2 + (-\frac{4\sqrt{3}}{3})^2 + (\frac{4\sqrt{6}}{3})^2} = \sqrt{0 + \frac{16}{3} + \frac{32}{3}} = \sqrt{\frac{48}{3}} = \sqrt{16} = 4$$

(b) Find the angle θ between $\mathbf{v}$ and $\mathbf{w}$.

$$\begin{aligned} \mathbf{v} \cdot \mathbf{w} &= (3, \sqrt{3}, 2\sqrt{6}) \cdot (0, -\frac{4\sqrt{3}}{3}, \frac{4\sqrt{6}}{3}) \\ &= 3 \cdot 0 + \sqrt{3} \cdot (-\frac{4\sqrt{3}}{3}) + 2\sqrt{6} \cdot \frac{4\sqrt{6}}{3} \\ &= 0 - 4 + 16 = 12 \end{aligned}$$

$$\cos \theta = \frac{\mathbf{v} \cdot \mathbf{w}}{\|\mathbf{v}\| \|\mathbf{w}\|} = \frac{12}{6 \cdot 4} = \frac{1}{2} \Rightarrow \theta = \frac{\pi}{3} \text{ or } 60^\circ$$

(c) Find a unit vector $\mathbf{u}$ which has the same direction as $\mathbf{v}$.

$$\mathbf{u} = \frac{\mathbf{v}}{\|\mathbf{v}\|} = \frac{(3, \sqrt{3}, 2\sqrt{6})}{6} = (\frac{1}{2}, \frac{\sqrt{3}}{6}, \frac{\sqrt{6}}{3})$$

(d) Find a number α so that $\alpha\mathbf{w} - \mathbf{v}$ is perpendicular to $\mathbf{v}$.

$$(\alpha\mathbf{w} - \mathbf{v}) \cdot \mathbf{v} = 0$$

$$\alpha\mathbf{w} \cdot \mathbf{v} - \|\mathbf{v}\|^2 = 0$$

$$\alpha = \frac{\|\mathbf{v}\|^2}{\mathbf{w} \cdot \mathbf{v}} = \frac{6^2}{12} = 3$$

Problem 2. (20 points)

(a) Find an 2×2 matrix $\mathbf{A}$ such that $\mathbf{A} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 2x - 2y \\ x + 5y \end{bmatrix}$.

$$\mathbf{A} = \begin{bmatrix} 2 & -2 \\ 1 & 5 \end{bmatrix}$$

(b) Find an 3×3 elimination matrix $\mathbf{E}$ that subtracts $5/2$ times of row 1 from row 3.

$$\mathbf{E} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -\frac{5}{2} & 0 & 1 \end{bmatrix}$$

(c) Find an 3×3 permutation matrix $\mathbf{P}$ that exchanges rows 3 and 2, then rows 1 and 3.

$$\begin{aligned} \mathbf{P} &= \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} \leftarrow \mathbf{P}_{31}\mathbf{P}_{32} \\ &= \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix} \end{aligned}$$

(d) Find the inverse of the matrix $\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ if it is invertible.

$$\begin{aligned} \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}^{-1} &= \frac{1}{1 \cdot 4 - 3 \cdot 2} \begin{bmatrix} 4 & -2 \\ -3 & 1 \end{bmatrix} \\ &= -\frac{1}{2} \begin{bmatrix} 4 & -2 \\ -3 & 1 \end{bmatrix} = \begin{bmatrix} -2 & 1 \\ \frac{3}{2} & -\frac{1}{2} \end{bmatrix} \end{aligned}$$

Problem 3. (22 points)

(a) Find the **LDU** factorization of $A = \begin{bmatrix} -1 & 1 & -3 \\ 2 & -4 & 1 \\ -3 & -3 & -4 \end{bmatrix}$

$$A = \begin{bmatrix} \boxed{-1} & 1 & -3 \\ \textcircled{2} & -4 & 1 \\ -3 & -3 & -4 \end{bmatrix} \xrightarrow{\ell_{21}=-2} \begin{bmatrix} -1 & 1 & -3 \\ 0 & -2 & -5 \\ -3 & -3 & -4 \end{bmatrix} \xrightarrow{\ell_{32}=3} \begin{bmatrix} -1 & 1 & -3 \\ 0 & \boxed{-2} & -5 \\ 0 & \boxed{-6} & 5 \end{bmatrix}$$

$$\xrightarrow{\ell_{32}=3} \begin{bmatrix} -1 & 1 & -3 \\ 0 & -2 & -5 \\ 0 & 0 & 20 \end{bmatrix}$$

$$\Rightarrow A = LDU = \underbrace{\begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 3 & 3 & 1 \end{bmatrix}}_L \underbrace{\begin{bmatrix} -1 & & \\ & -2 & \\ & & 20 \end{bmatrix}}_D \underbrace{\begin{bmatrix} 1 & -1 & 3 \\ 0 & 1 & \frac{5}{2} \\ 0 & & 1 \end{bmatrix}}_U$$

(b) Given $A = \begin{bmatrix} 1 & 1 & 2 \\ 1 & 2 & 2 \\ 2 & 2 & 3 \end{bmatrix}$, find A^{-1} using the Gaussian-Jordan Elimination.

$$[A|I] = \begin{bmatrix} \boxed{1} & 1 & 2 & | & 1 & 0 & 0 \\ \textcircled{1} & 2 & 2 & | & 0 & 1 & 0 \\ \textcircled{2} & 2 & 3 & | & 0 & 0 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 1 & \textcircled{2} & | & 1 & 0 & 0 \\ 0 & 1 & 0 & | & -1 & 1 & 0 \\ 0 & 0 & \boxed{-1} & | & -2 & 0 & 1 \end{bmatrix}$$

$$\rightarrow \begin{bmatrix} 1 & \textcircled{1} & 0 & | & -3 & 0 & 2 \\ 0 & \boxed{1} & 0 & | & -1 & 1 & 0 \\ 0 & 0 & -1 & | & -2 & 0 & 1 \end{bmatrix}$$

$$\rightarrow \begin{bmatrix} 1 & 0 & 0 & | & -2 & -1 & 2 \\ 0 & 1 & 0 & | & -1 & 1 & 0 \\ 0 & 0 & -1 & | & -2 & 0 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 0 & | & -2 & -1 & 2 \\ 0 & 1 & 0 & | & -1 & 1 & 0 \\ 0 & 0 & 1 & | & 2 & 0 & -1 \end{bmatrix} = [I|A^{-1}]$$

$$\Rightarrow A^{-1} = \begin{bmatrix} -2 & -1 & 2 \\ -1 & 1 & 0 \\ 2 & 0 & -1 \end{bmatrix}$$

Problem 4. (20 points) Let $A = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 2 & 4 & 8 & 10 \\ 3 & 6 & 7 & 10 \end{bmatrix}$.

(a) Reduce A to the reduced row echelon form R .

$$A = \begin{bmatrix} \boxed{1} & 2 & 3 & 4 \\ \boxed{2} & 4 & 8 & 10 \\ \boxed{3} & 6 & 7 & 10 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & 3 & 4 \\ 0 & 0 & \boxed{2} & 2 \\ 0 & 0 & \boxed{-2} & -2 \end{bmatrix} \xrightarrow{U} \begin{bmatrix} 1 & 2 & \boxed{3} & 4 \\ 0 & 0 & \boxed{2} & 2 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\rightarrow \begin{bmatrix} 1 & 2 & 0 & 1 \\ 0 & 0 & 2 & 2 \\ 0 & 0 & 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} \boxed{1} & 2 & 0 & 1 \\ 0 & 0 & \boxed{1} & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} = R$$

(b) Identify the pivot and free variables, and then find the rank of A .

Pivot variables: x_1, x_3

Free variables: x_2, x_4

$$\text{rank}(A) = 2$$

(c) Find the column space of A , $C(A)$.

$$C(A) = \left\{ c_1 \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} + c_2 \begin{bmatrix} 3 \\ 8 \\ 7 \end{bmatrix} \mid c_1, c_2 \text{ any real numbers} \right\}$$

(d) Find the null space of A , $N(A)$.

$$AX = 0 \Rightarrow RX = 0, \text{ i.e. } \begin{cases} x_1 + 2x_2 + x_4 = 0 \\ x_3 + x_4 = 0 \end{cases}$$

$$\text{i) Set } x_2 = 1, x_4 = 0 \Rightarrow \begin{cases} x_1 + 2 = 0 \\ x_3 = 0 \end{cases} \Rightarrow \begin{cases} x_1 = -2 \\ x_3 = 0 \end{cases} \Rightarrow s_1 = \begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \end{bmatrix}$$

$$\text{ii) Set } x_2 = 0, x_4 = 1 \Rightarrow \begin{cases} x_1 + 1 = 0 \\ x_3 + 1 = 0 \end{cases} \Rightarrow \begin{cases} x_1 = -1 \\ x_3 = -1 \end{cases} \Rightarrow s_2 = \begin{bmatrix} -1 \\ 0 \\ -1 \\ 1 \end{bmatrix}$$

$$\text{Thus } N(A) = \left\{ c_1 \begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \end{bmatrix} + c_2 \begin{bmatrix} -1 \\ 0 \\ -1 \\ 1 \end{bmatrix} \mid c_1, c_2 \text{ any real numbers} \right\}$$

Problem 5. (18 points) True or False (no explanation needed).

(a) If a matrix $\mathbf{A}$ is invertible, then $\mathbf{A}^2$ must be invertible.

True $(\mathbf{A}^2)^{-1} = (\mathbf{A}^{-1})^2$

(b) The set of all vectors (b_1, b_2, b_3) with $b_1 \geq b_2 \geq b_3$ is a subspace of $\mathbb{R}^3$.

False $(3, 2, 1) \rightarrow (-3, -2, -1)$

(c) If a subspace of the vector space $\mathbf{M}$ contains $\begin{bmatrix} 2 & 0 \\ 0 & 0 \end{bmatrix}$ and $\begin{bmatrix} 0 & 0 \\ 0 & -1 \end{bmatrix}$, then it must contain the identity matrix $\mathbf{I}$.

True $\frac{1}{2} \begin{bmatrix} 2 & 0 \\ 0 & 0 \end{bmatrix} + (-1) \begin{bmatrix} 0 & 0 \\ 0 & -1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

(d) If $\mathbf{A}$ and $\mathbf{B}$ are symmetric, then their product $\mathbf{AB}$ must be symmetric.

False $\begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} = \begin{bmatrix} 1 & -2 \\ 2 & -1 \end{bmatrix}$ not symmetric

(e) There exist two 2×2 matrices, $\mathbf{A}$ and $\mathbf{B}$, such that $\mathbf{A} \neq \mathbf{0}$ and $\mathbf{B} \neq \mathbf{0}$ but $\mathbf{AB} = \mathbf{0}$.

True $\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \cdot \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} = \mathbf{0}$

(f) The column space of $2\mathbf{A}$ equals the column space of $\mathbf{A}$ where $\mathbf{A}$ is a matrix.

True $C(2\mathbf{A}) = C(\mathbf{A})$ since $(2\mathbf{A})\mathbf{x} = 2(\mathbf{A}\mathbf{x})$